

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

B. TECH DEGREE SECOND SEMESTER SECOND INTERNAL EXAMINATIONS,
JULY 2024

23-200-0202B ENGINEERING CHEMISTRY

(CS A & B, EC B, IT A & B, EEE B)

(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

- CO1: Explain the basic concepts of chemical thermodynamics, and quantum chemistry.
- CO 2: Illustrate the spectroscopic methods in characterizing materials.
- CO3: Develop electro chemical methods to protect different metals from corrosion.
- CO4: Interpret the chemistry of a few important engineering materials and their industrial applications
- CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations

PART A

(Answer all questions)

(5 x 2= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
I (a)	Why TMS is used as reference standard in NMR spectroscopy?	2	L2	2	3
(b)	Explain the significance of fingerprint region.	2	L2	2	1
(c)	Explain how you can calculate the bond length of a diatomic molecule from its rotational spectrum.	2	L3	2	1
(d)	How polymers are classified based on their origin.?	2	L1	4	1
(e)	What is electrode potential?	2	L1	3	1

PART B

(Answer the following questions choosing either the first OR second option) (2 x 10= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II	(i) Which type of nuclei are NMR active? Under what conditions such nuclei will be in resonance with radiation absorbed?	6	L3	2	3
	The internuclear distance of CO is 1.13 Å.	4	L2	2	1

	Calculate the energy in Joules in the first excited rotational level. The atomic masses are $^{12}\text{C} = 1.99 \times 10^{-26} \text{ Kg}$ and $^{16}\text{O} = 2.66 \times 10^{-26} \text{ Kg}$				
OR					
III (a)	Explain the low resolution and high resolution NMR spectra of the following: (a) $\text{CH}_3\text{CH}_2\text{SH}$ (b) $\text{CF}_2=\text{CHF}$ (c) $\text{CH}_2(\text{CH}_2\text{CH}_2\text{Br})_2$	6	L3	2	3
(b)	State Beer – Lambert’s law. A solution of thickness 2cm transmits 40% incident light. Calculate the concentration of the solution given that $\epsilon = 6000 \text{ dm}^3/\text{mol}/\text{cm}$.	4	L2	2	1
IV (a)	Discuss different types of Lubricants and its properties.	6	L1	3	1
(b)	Write a note on Lead storage battery.	4	L1	4	1
OR					
V (a)	Explain the rusting of iron with the help of electrochemical theory of corrosion.	6	L1	4	1
(b)	Write a note on classification and properties of Refractories.	4	L1	3	1

Bloom’s Taxonomy Levels

L1 : 46.1 % L2: 30.7% L3: 23.2%